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JINA CLIP: Your CLIP Model Is Also Your Text Retriever

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Abstract

Contrastive Language-Image Pretraining (CLIP) is widely used to train models to align images and texts in a common embedding space by mapping them to fixed-sized vectors. These models are key to multimodal information retrieval and related tasks. However, CLIP models generally underperform in text-only tasks compared to specialized text models. This creates inefficiencies for information retrieval systems that keep separate embeddings and models for text-only and multimodal tasks. We propose a novel, multi-task contrastive training method to address this issue, which we use to train the `jina-clip-v1` model and achieve the state-of-the-art performance on both text-image and text-text retrieval tasks.

1. Introduction

Text-image contrastively trained models, such as CLIP (Radford et al., 2021), create an aligned representation space for images and texts by leveraging pairs of images and their corresponding captions. Similarly, text-text contrastively trained models, like `jina-embeddings-v2` (Günther et al., 2023), construct a representation space for semantically similar texts using pairs of related texts such as question/answer pairs, query/document pairs, or other text pairs with known semantic relationships.

Because image captions are typically very short, CLIP-style models trained with them only support short text context lengths. They struggle to capture the richer information in longer texts, and as a result, perform poorly on text-only tasks. Our empirical study (Table 1) demonstrates that OpenAI's CLIP underperforms in all text retrieval tasks. This poses problems for many applications that use larger text inputs, like text-image retrieval, multimodal retrieval augmented generation (Zhao et al., 2023) and image generation.

In this paper, we present and demonstrate the effectiveness of a novel approach to contrastive training with large-scale image-caption pairs and text pairs. We jointly optimize for representation alignment of both text-image and text-text pairs, enabling the model to perform well at both kinds of tasks. Due to the lack of available multimodal multi-target datasets (e.g. text-text-image triplets) we use different datasets for each class of task and jointly train for both.

The resulting model, `jina-clip-v1`, performs comparably to EVA-CLIP (Sun et al., 2023) on the cross-modal CLIP Benchmark[1], while the text encoder by itself performs as well as similar models on MTEB Benchmark tasks (Muennighoff et al., 2023).

2. Related Work

Contrastive learning for text embeddings is well-established for training models for text-based information retrieval, semantic textual similarity, text clustering, and re-ranking. Reimers & Gurevych (2019) propose a dual encoder architecture for pairwise text similarity training. Ni et al. (2022) demonstrate that the dual-encoder architecture scales efficiently. Wang et al. (2022) and Günther et al. (2023) develop multi-stage training methods incorporating *hard negatives*. Mohr et al. (2024) bring textual similarity scores directly into the training. Günther et al. (2023) and Chen et al. (2024) extend text embedding models' maximum input length to 8,192 tokens.

Contrastive text-image pre-training has become increasingly popular since Radford et al. (2021) proposed the CLIP (Contrastive Language-Image Pre-training) paradigm. Numerous follow-up studies have sought to improve text-image training. Zhai et al. (2022) introduce *locked image tuning* (LiT), which involves fixing the weights of a trained image encoder and training a text encoder to align with its image representations. Kossen et al. (2023) generalize the LiT paradigm to a more flexible *Three Tower* architecture. Zhai et al. (2023) propose a modified sigmoid loss function for contrastive learning, demonstrating better performance on

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[1]https://github.com/LAION-AI/CLIP_benchmark

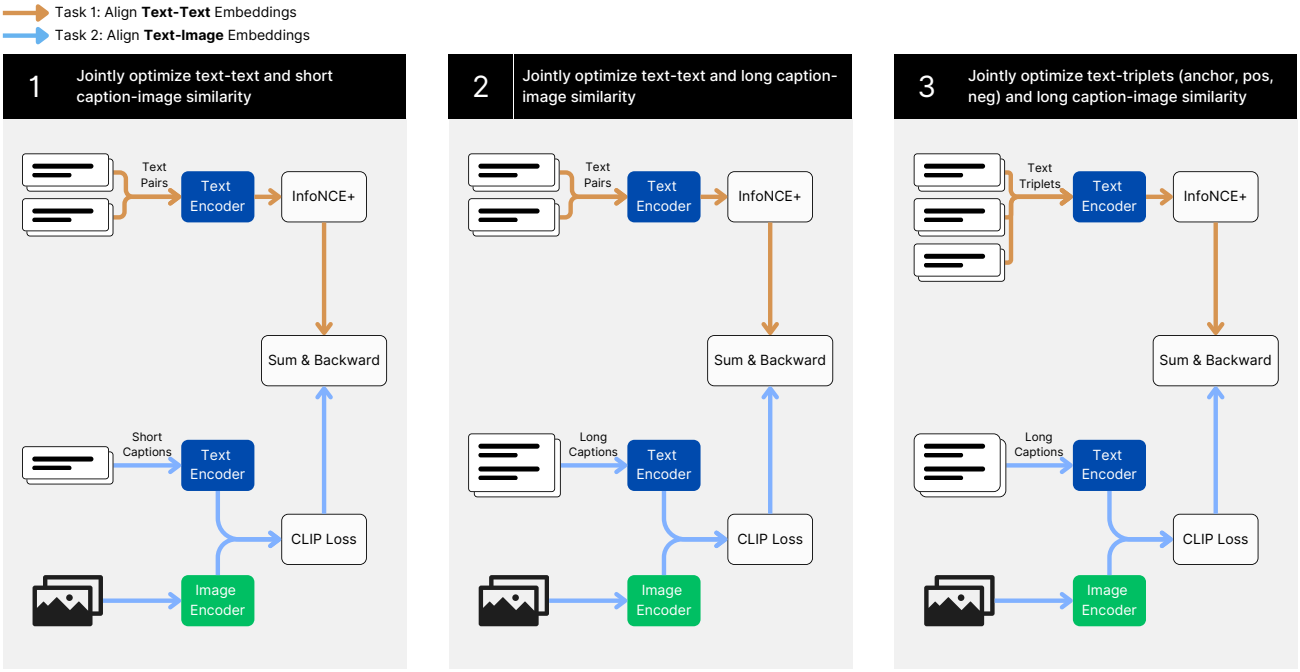

Figure 1. The training paradigm of `jina-clip-v1` , jointly optimizing text-image and text-text matching.

relatively small batch sizes. Cherti et al. (2023) and Sun et al. (2023) explore different setups for text-image training, including variations in datasets, model size, and hyperparameters. Zhang et al. (2024) empirically determine that the effective context length of CLIP is less than 20 tokens and propose an algorithm to stretch the positional encoding, improving performance on longer texts. Sun et al. (2024) scale up the EVA-CLIP architecture to 18B parameters.

Furthermore, a growing number of large datasets, such as YFCC100M (Thomee et al., 2016), LAION-5B (Schuhmann et al., 2022), and curated datasets like ShareGPT4v (Chen et al., 2023) help to constantly improve the performance of CLIP-like models.

3. Model Architecture

We use the same dual encoder architecture introduced in the original CLIP (Radford et al., 2021). It comprises a text encoder and an image encoder that generate representations of identical dimensionality.

The text encoder uses the JinaBERT architecture (Günther et al., 2023), a BERT variant that integrates AliBi (Press et al., 2021) to support longer texts. We pre-train the model using the *Masked Language Modeling* objective from the original BERT model (Devlin et al., 2019). Experimental results indicate that this yields superior final performance compared to starting from a text embedding model that has already been fully trained using contrastive learning.

For the image encoder, we use the EVA02 architecture (Fang et al., 2023). To keep the model size comparable to the text encoder, we select the base variant and initialize our model with the EVA02 pre-trained weights. Our experiments show that EVA02 significantly outperforms comparable image encoders like DinoV2 (Oquab et al., 2024) and ViT B/16 models from OpenCLIP (Ilharco et al., 2021).

4. Training

Figure 1 illustrates our multi-task, three-stage training approach. This method jointly optimizes the model to perform two tasks: text-image matching and text-text matching.

To achieve text retrieval performance on par with state-of-the-art text embedding models, we employ a multi-stage training paradigm, similar to what Wang et al. (2022) and Günther et al. (2023) propose. This paradigm introduces a second training stage, analogous to conventional *fine-tuning* stages, that makes use of small high-quality datasets with hard negatives to improve performance on embedding tasks.

We complement the two stages with text-image pairs for a two-task two-stage training pipeline. To address the short effective context length of CLIP-like models, we introduce an intermediate stage for training on long captions. Due to limited availability of public data, we rely on AI-generated long textual descriptions for this stage. The reasoning be-

hind opting for a separate stage is two-fold: a) because of the dataset size difference, integrating this data into the pre-training stage will limit its impact, and b) using the triplet fine-tuning stage for long caption training will limit the effect of hard negatives, as the model is not yet able to effectively process long contexts.

The result is a three-stage training process with two tasks in each stage:

- **Stage 1** focuses on learning to align image and text representations while minimizing losses in text-text performance. To this end, we train on large-scale and weakly supervised text-image and text-text pair datasets.
- **Stage 2** presents longer, synthetic image captions to the model while continuing to train with text-text pairs.
- **Stage 3** uses hard negatives to further improve the text encoder in separating relevant from irrelevant text. To maintain text-image alignment, we continue training on long image captions.

4.1. Data Preparation

Our text pair corpus $\mathbb{C}^{text}_{pairs}$ consists of data from a diverse collection of 40 text-pair datasets, similar to the corpus used in Günther et al. (2023).

For text-image training in Stage 1, we use LAION-400M (Schuhmann et al., 2021) as our corpus $\mathbb{C}^{img(s)}_{pairs}$. LAION-400M contains 400M image-text pairs derived from Common Crawl and is widely used for multimodal training.

In Stages 2 and 3, we use the ShareGPT4V (Chen et al., 2023) dataset as our $\mathbb{C}^{img(l)}_{pairs}$ corpus. This dataset contains approximately 100K synthetic captions generated with GPT4v (OpenAI, 2023) and an additional 1.1M long captions generated by a large captioning model trained on the original GPT4v generated output. This comes to a total of roughly 1.2M image captions. It would be interesting to investigate the impact of AI-generated data on performance, but that is outside the scope of this work.

Finally, in Stage 3, we use a triplet text corpus $\mathbb{C}^{text}_{triplets}$ that includes hard negatives. This corpus combines data from MSMarco (Bajaj et al., 2016), Natural Questions (NQ) (Kwiatkowski et al., 2019), HotpotQA (Yang et al., 2018) and the Natural Language Inference (NLI) dataset (Bowman et al., 2015). Each training batch contains one annotated positive and seven negative items. We select hard negatives using text retrieval models to emphasize relevance in text triplets, except for NLI where negatives are chosen randomly.

4.2. Loss Functions

All three stages employ a joint loss function that combines two InfoNCE loss functions (Van den Oord et al., 2018). For the text pairs in stage 1 and stage 2, we use the $\mathcal{L}_{\text{nce}}$ loss function of pairs of text embeddings $(\mathbf{q}, \mathbf{p}) \sim \mathbf{B}$ within a batch $\mathbf{B} \subset \mathbb{D}^{\text{pairs}}$. This function evaluates the cosine similarity $cos(\mathbf{q}, \mathbf{p})$ between a given query q and its corresponding target p, relative to the similarity of all other targets in the batch. We sum the loss in both directions to preserve the symmetry of similarity measures:

$$\begin{aligned}
&\mathcal{L}_{\text{nce}}(\mathbf{B}) := \mathcal{L}^{\longrightarrow}_{\text{nce}}(\mathbf{B}) + \mathcal{L}^{\longleftarrow}_{\text{nce}}(\mathbf{B}), \text{ with} \\
&\mathcal{L}^{\longrightarrow}_{\text{nce}}(\mathbf{B}) := \mathbb{E}_{(\mathbf{q},\mathbf{p})\sim\mathbf{B}}\left[-\ln \frac{e^{cos(\mathbf{q},\mathbf{p})/\tau}}{\sum_{i=1}^{k} e^{cos(\mathbf{q},\mathbf{p_i})/\tau}}\right] \\
&\mathcal{L}^{\longleftarrow}_{\text{nce}}(\mathbf{B}) := \mathbb{E}_{(\mathbf{q},\mathbf{p})\sim\mathbf{B}}\left[-\ln \frac{e^{cos(\mathbf{p},\mathbf{q})/\tau}}{\sum_{i=1}^{k} e^{cos(\mathbf{p},\mathbf{q_i})/\tau}}\right]
\end{aligned} \tag{1}$$

The constant temperature parameter τ influences how the loss function weighs minor differences in the similarity scores (Wang & Liu, 2021). In accordance with related work (Günther et al., 2023), we choose $\tau = 0.05$.

Similarly, we apply $\mathcal{L}_{\text{nce}}$ to pairs of caption and image embeddings $(\mathbf{c}, \mathbf{i}) \sim \mathbf{B}$ in batches $\mathbf{B} \subset \mathbb{D}^{\text{img}}$ to obtain loss values for text-image matching. For text-image training, τ is trainable, following the default behaviour in the OpenCLIP framework (Ilharco et al., 2021).

For text-text training in stage 3, we use text embeddings from the triplet database $(\mathbf{q}, \mathbf{p}, \mathbf{n_1}..., \mathbf{n_7}) \sim \mathbf{B}$ drawn in batches $\mathbf{B} \subset \mathbb{D}^{\text{triplets}}$. Recall that these consist of a query $\mathbf{q}$, a positive match $\mathbf{p}$, and seven negatives $\mathbf{n_1}..., \mathbf{n_7}$. We employ an extended version of the $\mathcal{L}_{\text{nce}}$ loss, denoted here as $\mathcal{L}_{\text{nce}^+}$, in Equation (2). Similar to $\mathcal{L}_{\text{nce}}$, this loss function is bidirectional but incorporates additional negatives when pairing queries with passages:

$$\begin{aligned}
&\mathcal{L}_{\text{nce}^+}(\mathbf{B}) := \\
&\quad \mathbb{E}_{r\sim\mathbf{B}}\left[-\ln \frac{e^{cos(\mathbf{q},\mathbf{p})/\tau}}{\sum_{i=1}^{k}\left[e^{cos(\mathbf{q},\mathbf{p_i})/\tau} + \sum_{j=1}^{7} e^{cos(\mathbf{q},\mathbf{n_{j,i}})/\tau}\right]}\right] \\
&+ \mathbb{E}_{r\sim\mathbf{B}}\left[-\ln \frac{e^{cos(\mathbf{p},\mathbf{q})/\tau}}{\sum_{i=1}^{k} e^{cos(\mathbf{p},\mathbf{q_i})/\tau}}\right] \\
&\text{with } r = (\mathbf{q}, \mathbf{p}, \mathbf{n_1}, \ldots, \mathbf{n_7}).
\end{aligned} \tag{2}$$

Table 1. Evaluation results on CLIP Benchmark and MTEB

Benchmark	**CLIP Benchmark**		**MTEB**			
Task Type	Zero-Shot Retrieval		Retrieval		STS	Avg MTEB Score
Model - Metric	txt-img r@5	img-txt r@5	r@5	ndcg@10	spearman	score
OpenAI CLIP ViT B/16	75.62	88.12	15.88	17.63	66.22	43.95
EVA-CLIP ViT B/16	**82.15**	90.59	22.92	26.03	69.62	47.64
LongCLIP ViT B/16	81.72	**90.79**	25.96	28.76	68.57	47.71
`jina-embeddings-v2`	-	-	42.56	47.85	80.70	**60.38**
`jina-clip-v1`stage 1	78.05	86.95	36.29	39.52	77.96	56.51
`jina-clip-v1`stage 2	81.86	90.59	36.80	40.44	78.33	57.19
`jina-clip-v1`	80.31	89.91	**43.05**	**48.33**	**80.92**	60.12

txt-img r@5 : Text to Image Recall@5 [%] img-txt r@5 : Image to Text Recall@5 [%] r@5 : Recall@5 [%]
spearman: Spearman Correlation

4.3. Training Steps

In each stage, the text and image encoders are applied to inputs from the corpora described in Section 4.1 and the training uses the following combinations of loss functions:

$$\begin{aligned}\mathcal{L}_1(\mathbf{B}_{text;s}, \mathbf{B}_{img;s}) &:= \mathcal{L}_{\text{nce}}(\mathbf{B}_{text;s}) + \mathcal{L}_{\text{nce}}(\mathbf{B}_{img;s}) \\ \mathcal{L}_2(\mathbf{B}_{text;l}, \mathbf{B}_{img;l}) &:= \mathcal{L}_{\text{nce}}(\mathbf{B}_{text;l}) + \mathcal{L}_{\text{nce}}(\mathbf{B}_{img;l}) \\ \mathcal{L}_3(\mathbf{B}_{text3}, \mathbf{B}_{img;l}) &:= \mathcal{L}_{\text{nce}}(\mathbf{B}_{text3}) + \mathcal{L}_{\text{nce}^+}(\mathbf{B}_{img;l})\end{aligned} \tag{3}$$

For stage 1, $\mathbf{B}_{text;s}$ is obtained from $\mathbb{C}^{text}_{pairs}$ by truncating the text values during tokenization to 77 tokens as in Radford et al. (2021). This enables us to use very large batches of size $32,768$. $\mathbf{B}_{img;s}$ is obtained from $\mathbb{C}^{img(s)}_{pairs}$ with the same truncation, albeit most captions in this corpus are short.

For stage 2, $\mathbb{C}^{text}_{pairs}$ is used again. However, text values are truncated to 512 tokens in this case, and as a result a smaller batch size of $8,192$ is used. The text image pairs $\mathbf{B}_{img;l}$ are selected from $\mathbb{C}^{img(l)}_{pairs}$. During this stage, text-text and text-image retrieval improves by adding synthetic data with longer captions to the training.

The last stage uses text triplets from $\mathbb{C}^{text}_{triplets}$ and the text-image batches $\mathbf{B}_{img;l}$ as in stage 2. This focused fine-tuning using text triplets and hard negatives brings text-text performance up to competitive levels with specialized text-only models. Training settings as well as training times for each stage are given in detail in Appendix table 2.

5. Evaluation

We evaluate our model's performance on text-only tasks, image-only tasks, and cross-modal tasks with both text and images. Table 1 shows the results of tests comparing `jina-clip-v1` to OpenAI CLIP (Radford et al., 2021), EVA-CLIP (Sun et al., 2023), and LongCLIP ViT B/16 (Zhang et al., 2024) models. Additionally, for text retrieval performance, we include a comparison with `jina-embeddings-v2`. These results demonstrate our model's high performance across all benchmarks.

To evaluate the model's cross-modal performance, we use the CLIP Benchmark which includes zero-shot image-classification and zero-shot cross-modal retrieval tasks. For zero-shot image-text and text-image information retrieval, we evaluate using Flickr8k (Hodosh et al., 2013), Flickr30K (Young et al., 2014) and MSCOCO Captions (Chen et al., 2015), which are all included in CLIP Benchmark. `jina-clip-v1` achieves an average Recall@5 of 85.8% across all retrieval benchmarks, outperforming OpenAI's CLIP model and performing on par with EVA-CLIP.

To evaluate `jina-clip-v1`'s text encoder, we use the Massive Text Embedding Benchmark (MTEB) (Muennighoff et al., 2023), which includes eight tasks involving 58 datasets. CLIP-like models generally perform poorly on text embedding tasks, particularly information retrieval. However, `jina-clip-v1` competes closely with top-tier text-only embedding models, achieving an average score of 60.12%. This improves on other CLIP models by roughly 15% overall and 22% in retrieval tasks. Detailed results are provided in the appendix.

6. Conclusion

We have presented a multi-task, three-stage training method that enables multimodal models to retain high levels of performance on text-only tasks. The model we produced using this method, `jina-clip-v1`, exhibits strong performance in cross-modal tasks like text-image retrieval and excels in tasks like semantic textual similarity and text retrieval. This result confirms that unified multimodal models can replace separate models for different task modalities, with large potential savings for applications. This model is currently limited to English-language texts due to limited multilingual resources. Future work will focus on extending this work to multilingual contexts.

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A. Appendix

Table 2. Training settings on each stage

Parameter	**Stage 1**	**Stage 2**	**Stage 3**
Image encoder weights init	EVA02 ViT B/16	Stage 1	Stage 2
Text encoder weights init.	JinaBERT v2	Stage 1	Stage 2
Peak learning rate	1e-4	5e-6	1e-6
Image-text pairs batch size	$32,768$	$8,192$	$1,024$
Text pairs batch size	$32,768$	$8,192$	$1,024$
Total steps	$60,000$	$1,500$	$7,000$
Max sequence length	77	512	512
Image-text pairs samples seen	2B	12M	7M
Text pairs samples seen	2B	12M	7M
Number of GPUs - H100s 80GB	8	8	8
Training time including evaluations	180h	3h	4h30m
Learning rate schedule	cosine decay		
Optimizer	AdamW (Loshchilov & Hutter, 2017)		
Optimizer hyper-parameters	$\beta_1, \beta_2, \epsilon = 0.9, 0.98, 1e-6$		
Weight decay	0.025		
Input resolution	(224, 224)		
Patch size	(16, 16)		
Numerical precision	AMP		

Table 3. Detailed performance on the CLIP Benchmark

Dataset - Model	JinaCLIP	JinaCLIP stage 1	JinaCLIP stage 2	OpenAI CLIP ViT B/16	EVA-CLIP ViT B/16	LongCLIP ViT B/16
Zero-shot Image Retrieval - Recall@5 [%]						
Average	80.31	78.05	81.86	75.62	**82.15**	81.72
Flickr30k	89.02	86.88	89.80	85.60	**91.10**	90.46
Flickr8k	85.50	84.18	87.26	82.84	**88.50**	88.40
MSCOCO	66.42	63.11	**68.54**	58.42	66.85	66.31
Zero-shot Text Retrieval - Recall@5 [%]						
Average	89.91	86.95	90.59	88.12	90.59	**90.79**
Flickr30k	96.50	93.80	96.10	96.20	96.60	**98.00**
Flickr8k	94.20	90.90	94.20	91.40	**94.60**	94.00
MSCOCO	79.02	76.14	**81.38**	76.76	80.58	80.38
Image Classification - Accuracy@1 [%]						
Average	43.28	46.74	45.39	46.16	**48.70**	46.67
Cars	68.03	76.89	69.39	64.73	**78.56**	59.17
Country211	13.45	15.69	13.68	**22.85**	21.34	20.28
Fer2013	**49.07**	38.45	47.55	46.18	51.17	47.80
Fgvc-aircraft	11.49	13.71	11.19	24.27	**25.11**	22.56
Gtsrb	38.70	41.93	39.77	43.58	**46.33**	42.93
Imagenet-a	29.92	33.20	30.68	49.93	**53.89**	46.84
Imagenet-o	33.40	32.40	34.00	42.25	34.10	**42.65**
Imagenet-r	73.66	76.07	74.00	77.69	**82.42**	76.63
Imagenet1k	59.08	64.16	59.81	68.32	**74.75**	66.84
Imagenet-sketch	45.04	49.33	45.90	48.25	**57.70**	47.12
Imagenetv2	51.37	55.71	52.21	61.95	**66.98**	60.17
Mnist	48.07	59.42	48.05	51.71	47.16	**71.84**
Objectnet	45.41	51.74	45.61	55.35	**62.29**	50.79
Renderedsst2	59.14	**60.90**	60.30	60.68	54.15	59.31
Stl10	97.89	98.19	97.96	98.28	**99.49**	98.41
Sun397	65.92	68.47	65.95	64.37	**70.62**	68.73
Voc2007	72.83	76.02	75.63	78.34	**80.17**	75.35
Voc2007-multilabel (mean-average-precision [%])	80.62	77.94	76.80	78.91	**83.08**	81.95
Vtab/caltech101	82.68	**84.58**	83.06	82.19	82.78	82.63
Vtab/cifar10	93.49	92.68	93.83	90.78	**98.46**	91.22
Vtab/cifar100	72.08	72.62	72.67	66.94	**87.72**	69.17
Vtab/clevr-closest-object-distance	15.61	**17.29**	15.45	15.83	15.72	15.90
Vtab/clevr-count-all	**22.35**	21.53	23.49	21.09	21.27	20.71
Vtab/diabetic-retinopathy	2.82	73.30	**73.47**	3.44	14.19	10.99
Vtab/dmlab	19.53	**21.51**	18.59	15.49	14.67	15.45
Vtab/dsprites-label-orientation	2.44	**3.33**	2.86	2.34	1.94	1.12
Vtab/dsprites-label-x-position	3.07	2.85	3.14	2.95	3.11	**3.15**
Vtab/dsprites-label-y-position	3.17	**3.28**	3.17	3.11	3.21	3.16
Vtab/dtd	55.43	**56.86**	55.11	44.89	52.82	45.27
Vtab/eurosat	49.52	47.00	48.35	55.93	**66.33**	60.44
Vtab/flowers	59.62	65.05	59.93	71.13	**75.75**	69.85
Vtab/kitti-closest-vehicle-distance	22.93	15.89	25.04	**26.44**	22.08	34.60
Vtab/pcam	55.54	**55.79**	53.30	50.72	50.95	52.55
Vtab/pets	80.98	86.97	80.59	89.04	**92.10**	89.21
Vtab/resisc45	55.46	57.89	54.67	58.27	60.37	**60.63**
Vtab/smallnorb-label-azimuth	**5.40**	5.09	5.14	5.21	4.96	5.14
Vtab/smallnorb-label-elevation	11.31	10.98	11.24	**12.17**	9.79	10.59
Vtab/svhn	25.46	22.47	24.55	31.20	17.65	**27.65**

Table 4. Performance of `jina-clip-v1`on MTEB Benchmark

Model	CF	CL	PC	RR	RT	STS	SM	Average
OpenAI CLIP ViT B/16	60.11	35.49	71.68	46.54	17.13	66.22	29.47	43.95
EVA-CLIP ViT B/16	60.96	37.67	74.91	47.91	25.41	69.62	28.39	47.64
LongCLIP ViT B/16	61.72	35.20	73.15	47.03	28.05	68.57	29.58	47.71
`jina-embeddings-v2`	**73.45**	41.74	**85.38**	**56.98**	47.85	80.70	31.60	**60.38**
`jina-clip-v1`stage 1	67.54	**44.57**	78.07	56.99	39.52	77.96	29.51	56.51
`jina-clip-v1`stage 2	69.45	43.76	80.03	57.26	40.44	78.33	29.09	57.19
`jina-clip-v1`	72.05	41.74	83.85	56.79	**48.33**	**80.92**	**30.49**	60.12

CF: Classification Accuracy [%] CL: Clustering $\mathcal{V}$ measure [%] PC: Pair Classification Average Precision [%]
RR: Reranking MAP [%] RT: Retrieval nDCG@10 STS: Sentence Similarity Spearman Correlation [%]
SM: Summarization Spearman Correlation [%]

Table 5. Detailed performance on the MTEB Classification tasks

Dataset - Model	Accuracy [%] JinaCLIP	Jina Embeddings-v2	JinaCLIP stage 1	JinaCLIP stage 2	OpenAI CLIP ViT B/16	EVA-CLIP ViT B/16	LongCLIP ViT B/16
Average Classification	72.05	**73.45**	67.54	69.45	60.11	60.96	61.72
AmazonCounterfactualClassification	68.16	**74.73**	59.85	60.78	59.58	60.92	60.76
AmazonPolarityClassification	**96.23**	88.54	93.23	95.95	63.42	63.32	64.26
AmazonReviewsClassification	44.54	**45.26**	42.26	43.25	29.39	31.33	31.65
Banking77Classification	83.94	**84.01**	82.82	83.25	73.31	74.42	74.79
EmotionClassification	47.07	**48.77**	41.16	41.24	34.58	32.65	37.11
ImdbClassification	91.75	79.44	86.02	**93.50**	58.66	57.29	57.53
MTOPDomainClassification	92.67	**95.68**	89.62	90.01	87.97	92.10	89.88
MTOPIntentClassification	64.58	**83.15**	58.74	60.44	63.36	65.76	65.98
MassiveIntentClassification	69.51	**71.93**	65.60	66.47	64.19	65.22	65.80
MassiveScenarioClassification	74.44	74.49	74.54	**74.82**	73.18	73.14	74.11
ToxicConversationsClassification	70.47	**73.35**	60.50	66.72	63.52	63.44	67.13
TweetSentimentExtractionClassification	61.22	**62.06**	56.15	56.97	50.12	51.96	51.70

Table 6. Detailed performance on the MTEB Clustering tasks

Dataset - Model	$\mathcal{V}$ measure JinaCLIP	Jina Embeddings-v2	JinaCLIP stage 1	JinaCLIP stage 2	OpenAI CLIP ViT B/16	EVA-CLIP ViT B/16	LongCLIP ViT B/16
Average Clustering	41.74	41.74	**44.57**	43.76	35.49	37.67	35.20
ArxivClusteringP2P	44.81	45.39	**46.26**	45.32	31.86	34.03	32.81
ArxivClusteringS2S	37.81	36.68	**39.55**	39.26	27.34	26.75	26.81
BiorxivClusteringP2P	34.74	37.05	**38.80**	36.20	31.27	31.03	30.07
BiorxivClusteringS2S	30.78	30.16	**34.53**	34.21	27.63	27.09	25.35
MedrxivClusteringP2P	30.82	32.41	**33.41**	31.54	29.27	29.36	30.30
MedrxivClusteringS2S	27.64	28.09	**31.54**	31.30	27.17	26.34	26.72
RedditClustering	56.21	53.05	**59.22**	59.09	42.94	49.94	42.94
RedditClusteringP2P	58.43	**60.31**	58.42	57.94	52.82	58.02	50.69
StackExchangeClustering	60.35	58.52	**64.16**	63.40	52.44	57.93	53.25
StackExchangeClusteringP2P	33.46	**34.96**	33.86	33.02	30.01	32.53	31.06
TwentyNewsgroupsClustering	44.08	42.47	**50.50**	50.12	37.61	41.33	37.18

Table 7. Detailed performance on the MTEB Pair-Classification tasks

Dataset - Model	Average precision based on cosine similarity						
	JinaCLIP	Jina Embeddings-v2	JinaCLIP stage 1	JinaCLIP stage 2	OpenAI CLIP ViT B/16	EVA-CLIP ViT B/16	LongCLIP ViT B/16
Average Pair Classification	83.85	**85.38**	78.07	80.03	71.68	74.91	73.15
SprintDuplicateQuestions	94.17	**95.30**	89.42	90.32	87.33	90.20	89.05
TwitterSemEval2015	71.18	**74.74**	62.08	66.39	53.04	55.36	55.21
TwitterURLCorpus	**86.20**	86.09	82.70	83.38	74.68	79.18	75.19

Table 8. Detailed performance on the MTEB ReRanking tasks

Dataset - Model	mAP@10						
	JinaCLIP	Jina Embeddings-v2	JinaCLIP stage 1	JinaCLIP stage 2	OpenAI CLIP ViT B1/6	EVA-CLIP ViT B/16	LongCLIP ViT B/16
Average Reranking	56.79	**56.98**	56.99	57.26	46.54	47.91	47.03
AskUbuntuDupQuestions	61.73	**62.25**	61.26	61.65	51.23	52.22	52.57
MindSmallReranking	31.21	30.54	31.42	**31.88**	26.42	28.00	26.93
SciDocsRR	81.76	83.10	**83.77**	83.58	71.05	70.80	70.61
StackOverflowDupQuestions	**52.47**	52.05	51.50	51.93	37.44	40.61	38.01

Table 9. Detailed performance on the MTEB Retrieval tasks

Dataset - Model	nDCG@10						
	JinaCLIP	Jina Embeddings-v2	JinaCLIP stage 1	JinaCLIP stage 2	OpenAI CLIP ViT B/16	EVA-CLIP ViT B/16	LongCLIP ViT B/16
Average Retrieval	**48.33**	47.85	39.52	40.44	17.13	25.41	28.05
ArguAna	**49.36**	44.18	39.53	48.26	15.51	23.49	32.01
ClimateFEVER	**24.81**	23.53	20.38	16.92	3.68	19.60	14.24
CQADupstackRetrieval	**40.92**	39.34	35.97	39.18	10.18	16.72	18.23
DBPedia	**36.64**	35.05	28.41	30.33	14.94	25.42	27.17
FEVER	**76.28**	72.33	57.50	46.72	33.45	59.26	63.54
FiQA2018	38.27	**41.58**	36.11	38.10	5.78	7.33	11.17
HotpotQA	**61.89**	61.38	40.24	43.87	9.30	21.54	33.61
MSMARCO	36.91	**40.92**	25.85	27.60	9.36	13.76	17.53
NFCorpus	**33.52**	32.45	31.65	32.17	16.44	21.83	27.21
NQ	58.09	**60.04**	40.07	41.23	5.28	10.89	21.20
QuoraRetrieval	87.88	**88.20**	81.55	84.32	76.63	82.32	78.31
SCIDOCS	**20.24**	19.86	20.06	20.20	3.46	7.40	9.24
SciFact	67.34	66.68	**68.77**	67.85	26.29	34.84	34.77
TRECCOVID	**71.61**	65.91	49.26	52.15	22.60	30.43	26.42
Touche2020	21.15	**26.24**	17.46	17.64	4.10	6.35	6.14

Table 10. Detailed performance on MTEB Retrieval tasks - Recall@5

Dataset - Model	Recall@5						
	JinaCLIP	Jina Embeddings-v2	JinaCLIP stage 1	JinaCLIP stage 2	OpenAI CLIP ViT B/16	EVA-CLIP ViT B/16	LongCLIP ViT B/16
Average - R@5	**43.05**	42.56	36.29	36.80	15.88	22.92	25.96
ArguAna	**62.37**	53.62	48.01	59.74	18.77	27.60	37.98
CQADupstackRetrieval	**44.80**	43.24	40.23	43.19	11.47	18.61	20.26
ClimateFEVER	**23.73**	22.26	19.80	16.33	3.38	18.57	13.33
DBPedia	**17.82**	16.61	15.37	15.71	6.78	11.42	12.62
FEVER	**85.93**	81.67	69.75	57.61	40.62	68.57	74.02
FiQA2018	38.18	**39.36**	34.80	36.36	5.83	7.69	11.41
HotpotQA	**58.95**	58.55	38.18	41.96	8.99	20.71	31.61
MSMARCO	46.16	**49.73**	32.57	34.04	11.73	16.85	21.59
NFCorpus	**13.04**	12.41	12.67	12.93	5.98	7.69	9.21
NQ	67.36	**70.37**	48.89	50.01	6.31	12.69	25.68
QuoraRetrieval	91.33	**91.69**	85.21	88.06	80.54	86.21	82.31
SCIDOCS	14.85	14.64	**14.86**	14.73	2.57	5.16	6.51
SciFact	72.11	73.27	**74.94**	73.79	33.08	38.89	40.34
TRECCOVID	**1.04**	1.01	0.74	0.78	0.32	0.47	0.46
Touche2020	8.04	**9.99**	8.39	6.79	1.79	2.67	2.10

Table 11. Detailed performance on the MTEB STS tasks

Dataset - Model	Spearman correlation based on cosine similarity						
	JinaCLIP	Jina Embeddings-v2	JinaCLIP stage 1	JinaCLIP stage 2	OpenAI CLIP ViT B/16	EVA-CLIP ViT B/16	LongCLIP ViT B/16
Average STS	**80.92**	80.70	77.96	78.33	66.22	69.62	68.57
BIOSSES	**83.75**	81.23	83.32	83.74	67.78	71.18	70.44
SICK-R	78.95	**79.65**	76.76	76.77	69.08	73.72	72.59
STS12	73.52	**74.27**	69.52	70.97	72.07	70.19	72.63
STS13	83.24	**84.18**	78.03	78.15	64.44	63.02	66.25
STS14	78.68	**78.81**	72.44	73.20	55.71	59.98	58.66
STS15	87.46	**87.55**	84.39	84.51	65.37	73.12	68.81
STS16	83.77	**85.35**	78.70	79.27	72.44	74.74	72.43
STS17	**89.77**	88.88	88.44	88.10	77.23	81.90	79.72
STS22	**65.15**	62.20	66.45	66.64	53.63	59.33	55.60
STSBenchmark	**84.93**	84.84	81.57	81.96	64.40	69.01	68.55

Table 12. Detailed performance on the MTEB Summarization tasks

Dataset - Model	Spearman correlation based on cosine similarity						
	JinaCLIP	Jina Embeddings-v2	JinaCLIP stage 1	JinaCLIP stage 2	OpenAI CLIP ViT B/16	EVA-CLIP ViT B/16	LongCLIP ViT B/16
SummEval	30.49	**31.60**	29.51	29.09	29.47	28.39	29.58